

CLOUDLESS.

Local AI, done properly.

Product & hardware strategy — the free OS, the roadmap, and two capital-light machines that turn a download into a business.

CloudlessOS — free

Cloudless Spark

Cloudless Mini

Build-to-order

Document

Product & Hardware Strategy

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Phase

0 — Orchestrator prototype

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01 PCs that are ready for local AI

Cloudless builds **PCs that are ready for local AI**, plus **CloudlessOS**, a Linux distro that makes installing and running local AI apps effortless. Your AI runs on your machine — for privacy, cost control, offline capability, and ownership.

The product is **not** a browser showing a web page. The kiosk shell is thin. The real product is three layers underneath it: an **orchestrator** that installs and runs AI apps, **hardware enablement** that makes a box "ready for local AI," and a **curated catalog** that makes one-click installs actually reliable.



Layer 1 — Orchestrator

A local daemon that installs, runs, updates and uninstalls AI apps as GPU containers; manages models, GPUs and networking.



Layer 2 — Hardware enablement

Drivers, CUDA/ROCm, VRAM detection, "recommend models that fit your machine." The hardest differentiator.



Layer 3 — Curated catalog

Vetted container recipes (ComfyUI, vLLM, Open WebUI...) so installs are reliable, not "install anything and pray."

Two products, one software core

CloudlessOS — the software

A free Linux distro. On a Cloudless PC it boots into a kiosk web UI — the machine *is* the appliance. Downloaded standalone, it opens in your normal browser. Same backend, only the shell differs.

Cloudless PC — the hardware

Local-AI-ready machines that ship with CloudlessOS pre-tuned. The free OS is the top-of-funnel; the configured machine is the revenue.

02 CloudlessOS — the engine of the funnel

Nobody pays for the OS. It is given away on purpose — it is the marketing engine that earns trust and creates demand for the machines. A free, excellent, standalone OS is the asset; the core is never paywalled.

What it does

- Turns the painful parts of local AI — CUDA/ROCm versions, Python/venv breakage, "which model fits my VRAM?", exposing services safely — into one-click, web-UI actions.
- Runs AI apps as **GPU containers** (Docker/Podman + NVIDIA Container Toolkit): reproducible, isolated, clean uninstall — unlike Pinokio's git-clone + conda approach.
- Manages models: download, dedupe, track disk, and recommend what fits the detected hardware.

Why "free OS" sells more hardware, not less

The same pattern is everywhere: **System76** gives Pop!_OS away and sells laptops; **Framework**, **Bazzite**, **Synology** — free or open software, paid hardware. Someone downloads CloudlessOS, runs it, sees it's genuinely effortless, then decides they'd rather buy the box where Cloudless already did the sourcing, tuning and testing. The free tier is the proof that makes the paid box an easy yes.

The model in one line

Free OS = the marketing engine. The pre-configured machine = the revenue. You never sell software — you sell **a box where it already works.**

03 Roadmap — software first, hardware last

Prove the magic moment on commodity hardware before committing to an OS image or building machines. Each phase de-risks the next.

Phase 0 — Orchestrator prototype

CURRENT

Goal — the magic moment: click an app in a web UI → it installs into a GPU container, downloads a sensible model, opens, and uninstalls cleanly.

- Dev env (WSL2 + Ubuntu, GPU visible in containers) ✓
- Daemon v0: install / run / stop / uninstall one app via a local API ✓
- Minimal web UI: catalog + install/launch + status
- Catalog of 3–4 apps (ComfyUI, vLLM, Open WebUI); model manager with "fits your VRAM"

Exit: a non-technical person installs and opens ComfyUI from the UI without touching a terminal.

Phase 1 — CloudlessOS image

NEXT

Package daemon + web UI + kiosk shell + drivers into a bootable distro. Choose the base (immutable Fedora-family vs Ubuntu), build a branded image, kiosk boot flow, onboarding, bootable ISO. **Publish the standalone download — the top-of-funnel goes live.**

Phase 2 — Cloudless PC (hardware)

LATER

Only once the software is the reason people want the machine. Reference hardware tuned for local AI, CloudlessOS pre-installed, plus the support and update story for shipped units.

The capital-light hardware ladder

A startup with no funding does not sign factory contracts. It climbs rungs, each funded by the previous one — exactly how Tiiny funded its device through pre-orders.

0

Buy one box. A single retail mini-PC (Spark or Strix Halo) is your dev box, demo unit, and the thing you film. Only required spend.

1

Ship the free OS. Build the audience. Costs bandwidth, nothing else.

2

Build-to-order. Customer pays first → you buy the retail unit with their money → flash, tune, test, drop-ship. No inventory, cash-positive per order.

3

Crowdfund the first real run. Pre-orders fund a small private-label batch — custom enclosure badge, boot logo, packaging. Customers' money pays the factory.

4

ODM white-label. Only with revenue and a proven campaign — e.g. Sixunited, AMD's authorized Strix Halo ODM. Walk in with traction, not a pitch.

04 Two machines, one OS

Both are **unified-memory appliances** — 128 GB of memory shared across CPU and accelerator, enough to hold large models on a low-power desktop box. Both start life as **retail units you configure and resell**, so neither needs funding to launch. They differ on one axis that changes everything: **CUDA or not**.

	Cloudless Spark	Cloudless Mini
Silicon	NVIDIA GB10 Grace-Blackwell (DGX Spark)	AMD Ryzen AI Max+ 395 (Strix Halo)
Memory	128 GB unified LPDDR5X	128 GB unified LPDDR5X-8000 (soldered)
Compute	Grace ARM CPU + Blackwell GPU, full CUDA	16× Zen 5, RDNA 3.5 iGPU (40 CU), 50+ TOPS NPU
Software	Full CUDA — your catalog runs natively	x86 Linux — ROCm / Vulkan / llama.cpp
Architecture	ARM64 (arm64 CUDA containers)	x86-64 (standard Linux ecosystem)
Storage	1–2 TB PCIe Gen4 NVMe	1–2 TB PCIe Gen4 NVMe
Retail base cost	~\$3,500–4,000	~\$1,800–2,000
Porting cost to you	~ zero (CUDA native)	High (validate catalog off-CUDA)
Long-term path	Reseller / integrator (NVIDIA gates GB10 silicon)	Can manufacture later via authorized ODM
Best for	Developers, maximum compatibility	Value buyers, the own-brand future

Cloudless Spark

GB10 · 128 GB · CUDA

The fastest path to revenue. Because it is CUDA, the entire existing software stack — vLLM, ComfyUI, the container catalog — runs with no porting. Buyable retail today as NVIDIA DGX Spark and OEM clones (Dell, ASUS, Gigabyte, Lenovo, MSI).

Trade-off: ~2× the price, and ARM64 (a few ecosystem packages still lack arm64 builds — test before promising). You remain a reseller: NVIDIA will not sell you GB10 to build your own board.

Cloudless Mini

Strix Halo · 128 GB · x86

The value machine and the only path to owning the metal. x86 keeps the whole Linux/container world working; ~\$2k retail base leaves healthier margin. Sixunited is AMD's authorized Strix Halo ODM, so a future private build is real.

Trade-off: not native CUDA — the catalog must be validated against ROCm / Vulkan / llama.cpp. That porting effort is the true cost of this path, and it dwarfs the sourcing.

You don't have to choose yet. CloudlessOS is platform-agnostic and build-to-order carries no inventory, so offer both and let the first 10–20 orders tell you which the market wants. Recommended start: **Spark for revenue now** (zero porting), **Mini as the "someday we build our own" track.**

Also on the bench: a halo **GPU-tower** line (2× RTX 3090 NVLink "Cloudless One", 2× RTX 5090 "Studio") for users who need fast, full-quality image/video generation an NPU can't match — documented separately.

05 The storefront — two buttons

The whole business fits on one web page a solo founder can stand up this week: a free download that builds the audience, and a build-to-order machine that turns that audience into cash — with no inventory and no funding.

① Download CloudlessOS — free

The ISO / image. Costs bandwidth, nothing else. This is the funnel: trust, audience, and proof the experience is good.

② Buy a Cloudless machine

A normal product page (Stripe / Shopify). Customer pays in full → you buy the retail unit → flash, tune, test, box, ship. Cash-positive per order.

What the buyer actually pays the premium for

- Hardware they didn't have to research or source.
- Pre-tuned and tested — "it just works" on that exact silicon (Layer 2).
- A curated model/app catalog ready on first boot.
- Support + warranty pass-through — one throat to choke.
- The appliance experience: plug in, open a browser, done.

Two honesty rules

- **Warranty split:** hardware RMA goes to the manufacturer; Cloudless supports CloudlessOS. Be explicit; don't imply you built the metal.
- **Origin:** "Powered by CloudlessOS, built on NVIDIA GB10 / AMD Strix Halo" is honest and still strong. Never misrepresent.

Later — park it, don't build it now

Once there's volume, optional recurring revenue can layer on (premium support tiers, encrypted off-site backup, a pro catalog). The core OS stays free, always. Don't compromise the asset that drives every machine sale.